

# Day 4 — Technical Logic

## Validation & Finalization

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**Activity packet** for participant quads. Today's job: catch every inconsistency across the artifact set, add the **SEP-light** and **DP-light**, integrate everything, and rehearse the presentation. Tomorrow you present.

### Where we are in the week

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The 4 documents from Day 3 (PRD-light, TCD-light, TMD-light, AI Spec v1) need to be **internally consistent** before they ship. Today's discipline is **technical logic validation** — checking that the data model supports the API, the SLOs match the architecture, the metrics align with the outcome map, and the trade-offs don't contradict.

Plus today produces the last 2 documents: **SEP-light** and **DP-light**, in compressed form.

### Inputs

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- All 4 documents from Day 3
- Day-3 cross-quad findings
- The SEP / DP templates from Weeks 6–7

### What "technical logic validation" means

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A common rookie failure: each artifact is fine in isolation, but together they don't fit. Examples:

- The PRD's NFR says "p95 < 500ms" but the TMD's sequence diagram totals 800ms
- The TCD says "modular monolith" but the TMD shows components that need to deploy independently
- The data model has no foreign key for a relationship the API requires

- The outcome map's "leading indicator" isn't measurable with the observability NFR

Technical logic validation = **walk every cross-artifact reference** and verify they're consistent.

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## The 7-point validation checklist

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| # | Check                                         | Where to look                                                |
|---|-----------------------------------------------|--------------------------------------------------------------|
| 1 | Outcome map → metrics → SLOs aligned          | DP-light Section 1, PRD-light Section 7, TCD-light Section 4 |
| 2 | Data model supports the API contract          | TMD-light Section 1 ↔ Section 3                              |
| 3 | API contract supports the AC                  | TMD-light Section 3 ↔ PRD-light Section 6                    |
| 4 | SLO budget fits the sequence diagram          | TCD-light Section 4 ↔ TMD-light Section 4                    |
| 5 | NFRs don't contradict the architecture stance | PRD-light Section 7 ↔ TCD-light Section 1                    |
| 6 | Trade-offs don't contradict each other        | TCD-light Section 5                                          |
| 7 | Stakeholder sign-off captures all constraints | TCD-light Section 6                                          |

A quad that passes all 7 has a coherent artifact set. Most fail 2–3 on first pass — that's expected; that's why this day exists.

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## Activity 1 — The 7-Point Validation

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**Format:** Quad • 40 min • Block 1

### Purpose

Walk every check; document what passes and what doesn't.

### Quad protocol

For each of the 7 checks, the quad walks the cross-references:

1. **Pull both sides of the check** (e.g., for #4: latency SLO target + sequence diagram totals)

2. **Compare** — do they agree?

3. **If yes**: mark pass.

4. **If no**: document the gap. Decide:

- Tighten the artifact in question (often the answer)
- Adjust the SLO / NFR (only if evidence demands)
- Mark as a known gap to call out in the presentation

The validation log:

```
| # | Check | Status | Gap / fix |
|---|-----|-----|-----|
| 1 | Outcome → metrics → SLOs | Pass | None |
| 2 | Data model ↔ API | Partial | Missing FK on
User → Order; added |
| 3 | API ↔ AC | Pass | None |
| 4 | SLO budget ↔ sequence | Fail → Fixed |
Sequence totaled 800ms; SLO was 500ms; tightened SLO
to p95 < 1000ms with defense |
| 5 | NFRs ↔ architecture | Pass | None |
| 6 | Trade-offs internal consistency | Pass | None
|
| 7 | Stakeholders ↔ constraints | Partial |
Compliance not on sign-off list; added |
```

## Output

A validation log + updated artifacts.

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## Activity 2 — SEP-Light + DP-Light

**Format:** Quad • 40 min • Block 2

### Purpose

Compressed Weeks 6 + 7 in 40 minutes. SEP-light captures stakeholder + 1 trade-off brief + 1 simulated negotiation; DP-light captures outcome map + backlog skeleton + 1 experiment.

## SEP-light template

```
# SEP-light – <feature>

## 1. Stakeholder map (compressed)
3 high-power stakeholders. Power × Interest
placement + RACI for 1 decision (the most
consequential).

## 2. 1 Trade-off brief
1 page. Pick one TCD-light Section 5 trade-off;
brief for the relevant stakeholder.

## 3. 1 Simulated negotiation outcome
For one of the 5-most-consequential asks, document:
– The ask
– Anticipated stakeholder response (top 1 objection
+ your response)
– Anticipated agreement / deferral / escalation path
```

## DP-light template

```
# DP-light – <feature>

## 1. Outcome map (compressed)
Primary outcome + 2 supporting outcomes + 1 counter-
outcome.

## 2. Backlog skeleton
1 Epic + 1 Feature + 5 User Stories (top-priority
subset of PRD-light Section 6).

## 3. Tracking plan (compressed)
3 leading indicators. 1 monthly review cadence.

## 4. Value stream (compressed)
Top 3 queues identified.

## 5. 1 Bottleneck removal experiment
Hypothesis / test / success criterion for one queue.
```

## Quad protocol

1. **SEP-light** (20 min):
  - 5 min: 3 stakeholders + RACI for the most consequential decision

- 10 min: 1 trade-off brief (pull from TCD-light Section 5)
  - 5 min: 1 simulated negotiation outcome
2. **DP-light** (20 min):
- 5 min: outcome map (primary + 2 supporting + counter)
  - 5 min: backlog skeleton
  - 5 min: 3 leading indicators
  - 5 min: 1 bottleneck experiment

### What "good" looks like

- SEP-light's 1 trade-off brief is in the **stakeholder's currency** (not technical)
- DP-light's primary outcome is **user-visible**
- The leading indicators are **measurable within 7 days**
- The bottleneck experiment has a **measurable success criterion**

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## Activity 3 — Cross-Artifact Integration Pass

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**Format:** Quad • 40 min • Block 3

### Purpose

Read all 6 documents top-to-bottom. Catch any remaining contradictions. Add forward / backward references between artifacts.

### The 6 documents

1. PRD-light
2. TCD-light
3. TMD-light
4. SEP-light
5. DP-light
6. AI Spec v1

### Integration check

For every cross-artifact reference, confirm:

- The reference is **specific** (cite Section, not "see the TCD")

- The referenced section **exists** and **says what's claimed**
- The reference **adds value** (linking provides context the reader needs)

### Quad protocol

1. **Solo read-through** (15 min). Each member reads all 6 documents.
2. **Pool issues** (10 min).
3. **Fix in priority order** (15 min): coherence first, prose second.

### What "good" looks like

- Every cross-reference is verified
- The 6 documents read as a coherent set, not 6 standalone pieces
- The AI Spec correctly synthesizes the others (the synthesis test)

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## Activity 4 — Presentation Rehearsal

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**Format:** Quad • 45 min + Wrap • Block 4

### Purpose

Rehearse the 15-minute Friday presentation. Time it. Get feedback. Fix what doesn't land.

## The presentation structure (15 min)

| Time  | Section                                  | What's covered                               |
|-------|------------------------------------------|----------------------------------------------|
| 1 min | Customer + problem                       | Real customer; specific friction             |
| 2 min | Outcome                                  | Primary outcome; tied to user/business value |
| 3 min | Architecture stance + top trade-off      | Defended in business terms                   |
| 3 min | Data + API + sequence highlights         | Engineering-ready summary                    |
| 2 min | SLO + key constraint                     | Latency / availability / one compliance item |
| 2 min | Stakeholder + delivery readiness         | SEP-light + DP-light highlights              |
| 2 min | What you'd do differently with more time | Honesty + reflection                         |

Plus 5 min Q&A.

## Quad protocol

1. **Decide who speaks** (5 min). All 3 quad members on stage; assign sections.
2. **Run-through #1 with timer** (15 min). One pass; nobody interrupts.
3. **Capture what didn't work** (10 min). Where did the timer break? Where did a section drag?
4. **Run-through #2** (15 min). Apply fixes. Time again.

## What "good" looks like

- The 15-min limit is held (within 30 sec)
- All three quad voices speak
- The customer + problem section grounds the rest
- The trade-off section is **defended in business terms**, not technical jargon
- The "what we'd do differently" section is **honest**, not performative

## Wrap (last 15 min)

Each quad shares:

- Their **opening sentence**
  - The **section they're least sure of**
  - One **anticipated Q&A question**
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## End-of-day checkpoint

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Each quad ends Day 4 with:

- ✓ **Validation log** — all 7 checks marked
- ✓ **SEP-light** (1 page)
- ✓ **DP-light** (1 page)
- ✓ **AI Spec v2** (after integration pass)
- ✓ **Final 6-document set**, integrated
- ✓ **Presentation rehearsed** at least twice with timer
- ✓ Final provenance log entry